FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - II

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1 A spherical object of mass 1 kg and radius 1 m is falling vertically downward inside a viscous liquid in a gravity free space. At a certain instant the velocity of the sphere is 2 m/s. If the coefficient of viscosity of the liquid is $\frac{1}{18\pi}$ N-S/m^2, then velocity of ball will become 0.5 m/s after a time

(A) ln 4 s
(B) 2 ln 4 s
(C) 3 ln 4 s
(D) 2 ln 2 s

2 Two radioactive elements R and S disintegrate as

$R \longrightarrow P + \alpha$; $\lambda_R = 4.5 \times 10^{-3}$ years $^{-1}$

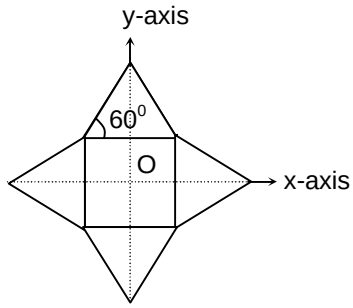
$S \longrightarrow Q + \beta$; $\lambda_S = 3 \times 10^{-3}$ years $^{-1}$

Starting with number of atoms of R and S in the ratio of 2 : 1, this ratio after the lapse of three half lives of R will be

(A) 3 : 2
(B) 1 : 3
(C) 1 : 1
(D) 2 : 1

3 Moment of inertia of a uniform symmetric plate as shown in figure about x-axis is I. Moment of inertia of this plate about an axis passing through centre of plate O and perpendicular to the plane of plate is

(A) 2I
(B) I
(C) I/2
(D) I/4



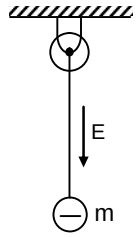
Space for Rough work

4. A body (solid sphere) of mass m makes an elastic collision with another identical body at rest. Just after collision the angle between the velocity vector of one body with the initial line of motion is 15° then the angle between velocity vector of the other body with the initial line of motion is

(A) 75^0 (B) 60^0

(C) 45^0 (D) 30^0

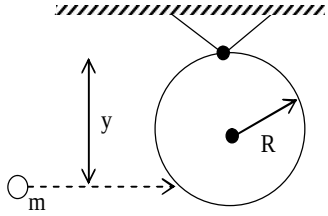
5. A simple pendulum charged negatively to q coulomb oscillates with a time period T in a downward electric field. If the electric field is with-drawn, the new time period is:



(A) = T (B) > T

(C) < T (D) $T\frac{mg}{qE}$

6. A small ball of mass m hits the cylinder which is hinged at top and free to rotate in vertical plane as shown in figure. Mass of cylinder is M. Liner momentum of system (ball + cylinder)



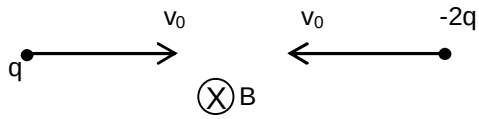
(A) Remains conserved if y = R

(B) Remains conserved if $y = \frac{3}{2}R$

(C) Remains conserved if $y = \frac{7}{5}R$

(D) Cannot be conserved for any value of y.

7. A charge particle q of mass m is placed at a distance d from another charge particle – 2q of mass 2m in a uniform magnetic field B as shown. If particles are projected towards each other with equal speed v_0, so that the two particles touches each other without collision during its motion. (Assume only force due to magnetic field acts on the particle)

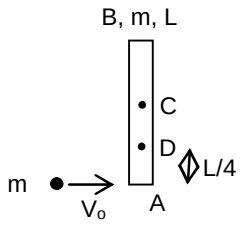


(A) $\frac{qBd}{m}$ (B) $\frac{qBd}{2m}$

(C) $\frac{2qBd}{m}$ (D) $\frac{3qBd}{2m}$

Space for Rough work

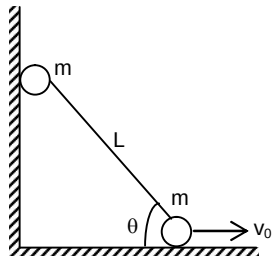
8. A thin, uniform rod of mass M and length L is at rest on a smooth horizontal surface. A particle of same mass M collides with the rod at one end perpendicular to its length with velocity V_o and sticks to it. C is the middle point of the rod and D is a point at a distance of L/4 from end A. The time taken by the rod turn through 90° after collisions



(A) $\frac{L}{V_o}$ (B) $\frac{\pi L}{V_o}$

(C) $\frac{3\pi L}{5V_o}$ (D) $\frac{5\pi L}{12V_o}$

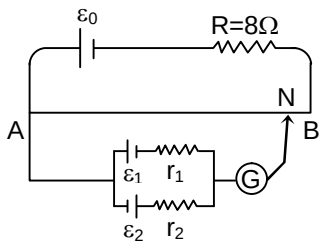
9. In the shown diagram two point masses m are joined by a massless rod of length 5.5 m. The lower end of the mass system is dragged with constant velocity v_0 rightwards. The magnitude of velocity of centre of mass when $\theta = 37^0$, will be



(A) $\frac{5v_0}{6}$ (B) $2v_0$

(C) $v_0\sqrt{2}$ (D) $\frac{6v_0}{5}$

10. A battery of emf $\varepsilon_0 = 12V$ is connected across a 4m long uniform wire having resistance 4Ω/m. The cells of small emfs $\varepsilon_1 = 2V$ and $\varepsilon_2 = 4V$ having internal resistance 2Ω and 6Ω respectively are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to :



(A) $\frac{1}{6}$m (B) $\frac{1}{3}$m

(C) 25 cm (D) 50 cm

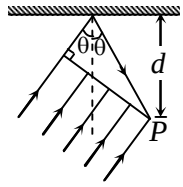
Space for Rough work

11. A uniform rod of length L and mass M is pivoted freely at one end and released as shown. The angular acceleration of the rod and the tangential linear acceleration of the free end when the rod is horizontal are
(A) 3g/2L, 3g/2 (B) 3g/2L, g
(C) g/2L, g/2 (D) 6g/L, 6g

12. A wheel of mass 2 kg and radius 10 cm is rotating about its axis at an angular velocity of 2π rad/s. The force that must be applied tangentially to the wheel to stop it in 5 revolutions is
(A) $3\pi\times10^{-2}$ kg-m^2/s (B) $4\pi \times 10^{-2}$ kg-m^2/s
(C) $\pi\times 10^{-2}$ kg-m^2/s (D) $2\pi\times 10^{-2}$ kg-m^2/s

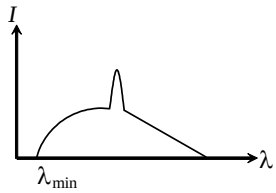
13. When a hydrogen atom emits a photon in going from n = 5 to n = 1 state, its recoil speed is almost (mass of H-atom = 1.67×10^{-27} kg)
(A) 10^{-4} m/s (B) 2×10^{-2} m/s
(C) 4 m/s (D) 8×10^{-2} m/s

14. A plane wave front of light is incident on a plane mirror. Intensity is maximum at P when



(A) $\cos\theta = \frac{\lambda}{2d}$ (B) $\cos\theta = \frac{3\lambda}{4d}$
(C) $\sec\theta - \cos\theta = \frac{3\lambda}{4d}$ (D) $\sec\theta - \cos\theta = \frac{\lambda}{2d}$

15. An X-ray tube has three main controls.
(i) the target material (its atomic number Z)
(ii) the filament current (I_f) and
(iii) the accelerating voltage (V)
Figure shows a typical intensity distribution against wavelength.



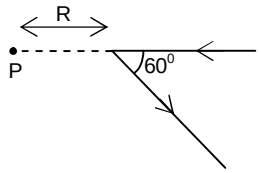
Which of the following is incorrect?
(A) The limit λ_{min} is proportional to V^{-1}
(B) The sharp peak shifts to the right as Z is increased
(C) The penetrating power of X ray increases if V is increased
(D) The intensity everywhere increases if filament current I_f is increased

Space for Rough work

16. The image of an object, formed by a plano-convex lens at a distance of 8 m behind the lens, is real and is one-third the size of the object. The wavelength of light inside the lens is $\frac{2}{3}$ times the wavelength in free space. The radius of the curved surface of the lens is

(A) 1 m (B) 2 m
(C) 3 m (D) 6 m

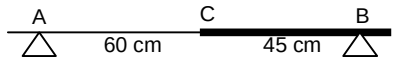
17. A long straight wire, carrying a current I is bent at its mid point to form an angle of 60^0. AT a point P, distance R from the point of bending the magnetic field is



(A) $\frac{\left(\sqrt{2}-1\right)\mu_0 i}{4\pi R}$ (B) $\frac{\left(\sqrt{2}+1\right)\mu_0 i}{4\pi R}$

(C) $\frac{\mu_0 i}{4\sqrt{3}\pi R}$ (D) $\frac{\mu_0 i}{8R}$

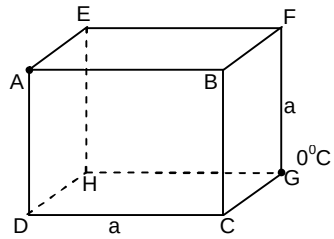
18. A steel wire of length 60 cm and area of cross section $10^{-6}m^2$ is joined with an aluminium wire of length 45 cm and area of cross section $3\times10^{-6}m^2$. The composite string is stretched by a tension of 80 N. Density of steel is 7800 kg m^{-3} and that of aluminium is 2600 kg m^{-3}. The minimum frequency of turning fork, which can produce standing wave in it with node at joint is



(A) 357.3 Hz (B) 375.3 Hz
(C) 337.5 Hz (D) 325.3 Hz

Space for Rough work

19. A cubical frame is made by connecting 12 identical uniform conducting rods as shown in the figure. In the steady state the temperature of junction A is 100^0 C while that of the G is 0^0C. Then,



(A) B will be Hotter than H

(B) Temperature of F is 40^0C

(C) Temperature of D is 66.67^0C

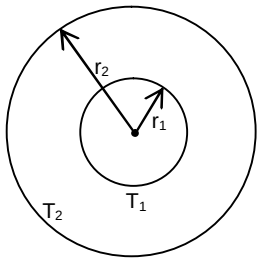
(D) Temperature of E is 50^0C

20. A body of mass 1kg is suspended from a massless spring having force constant 600 N/m. Another body of mass 0.5 kg moving vertically upwards hits the suspended body with a velocity 3 m/sec and gets embedded in it. The frequency of oscillation and the amplitude of motion are

(A) $\frac{10}{\pi}$Hz, 10Cm (B) $\frac{10}{\pi}$Hz, 5Cm

(C) $\frac{5}{\pi}$Hz, 5Cm (D) $\frac{5}{\pi}$Hz, 10Cm

21. The figure shows a hollow conducting sphere radius (r_2) with cavity radius (r_1) kept at temperatures T_1 and T_2 respectively. The radial rate of flow of heat in a substance between the two concentric spheres in proportional to



(A) $\frac{r_1 r_2}{r_2 - r_1}$ (B) $(r_2 - r_1)$

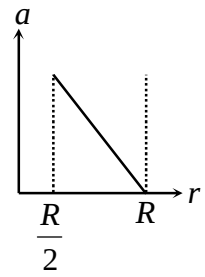
(C) $\frac{(r_2 - r_1)}{r_1 r_2}$ (D) $\log_e\left(\frac{r_2}{r_1}\right)$

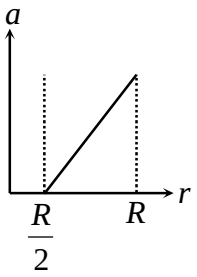
Space for Rough work

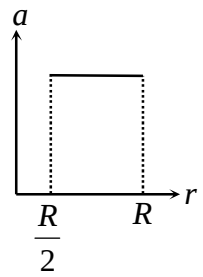
22. Two spherical bodies of masses M and 5M and radii R and 2R respectively are released in free space with initial separation between their centres equal to 12R. If they attract each other due to gravitational force only, then the distance covered by smaller body just before collision is

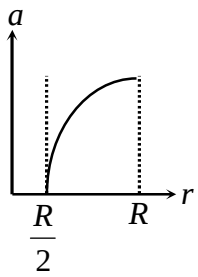
(A) 1.5 R (B) 2.5 R
(C) 4.5 R (D) 7.5 R

23. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $\frac{R}{2}$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is

(A) 

(B) 

(C) 

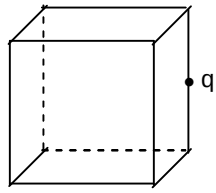
(D) 

Space for Rough work

24. Two longitudinal waves propagating in the X and Y directions superimpose. The wave equations are as below $\psi_1 = A\cos(\omega t - kx)$ and $\psi_2 = A\cos(\omega t - ky)$. Trajectory of the motion of a particle lying on the line $y = x + \frac{(2n+1)\lambda}{2}$ will be

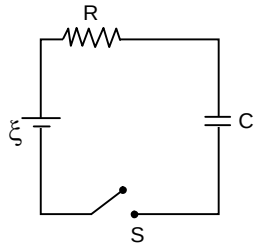
(A) straight line (B) circle
(C) ellipse (D) none of these

25. The total flux of electric lines of forces of a charge q placed at the mid point of the edge of a side of rectangular box, as shown in figure through the box is



(A) $\frac{q}{2\epsilon_o}$ (B) $\frac{q}{4\epsilon_o}$
(C) $\frac{q}{8\epsilon_o}$ (D) $\frac{q}{16\epsilon_o}$

26. As situation shown in figure the maximum value of rate of energy stored in the capacitor after the switch is closed



(A) $\frac{\xi^2}{2R}$ (B) $\frac{\xi^2}{4R}$
(C) $\frac{\xi^2}{8R}$ (D) none of these

27. If vectors P, Q and R have magnitude 5, 12 and 13 units and $\vec{P} + \vec{Q} = \vec{R}$, the angle between Q and R is

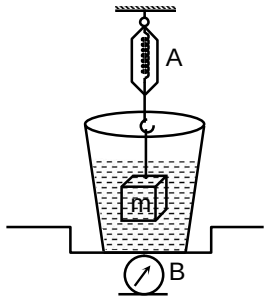
(A) $\cos^{-1}\frac{5}{12}$ (B) $\cos^{-1}\frac{5}{13}$
(C) $\cos^{-1}\frac{12}{13}$ (D) $\cos^{-1}\frac{7}{13}$

Space for Rough work

28. A particle is projected from ground At a height of 0.4 m from the ground, the velocity of a projectile in vector form is $\vec{v} = (6\hat{i} + 2\hat{j})$ m/s (the x-axis is horizontal and y-axis is vertically upwards). The angle of projection is (g = 10 m/s^2):
(A) 45° (B) 60°
(C) 30° (D) $\tan^{-1}(3/4)$

29. Force acting on a particle moving in a straight line varies with the velocity of the particle as $F = \frac{K}{v}$. Here, K is a constant. The work done by the this force in time t is:
(A) $\frac{K}{v^2}.t$ (B) 2Kt
(C) Kt (D) $\frac{2Kt}{v^2}$

30. The spring balance A reads 2 kg with a block m suspended from it. A balance B reads 5 kg when a beaker filled with liquid is put on the pan of the balance. The two balances are now so arranged that the hanging mass is inside the liquid as shown in figure. In this situation
(A) The balance A will read more than 2 kg.
(B) The balance B will read less than 5 kg.
(C) The balance A will read less than 2 kg and B will read more than 5 kg.
(D) The balances A and B will read 2 kg and 5 kg respectively



Space for Rough work

Chemistry PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. How many electron(s) is/are present in 1.8 mL of water (density = 1 g/mL)?
(A) $0.1 \times 6.023 \times 10^{23}$ (B) $10 \times 6.023 \times 10^{23}$
(C) 6.023×10^{23} (D) $1.8 \times 10 \times 6.023 \times 10^{23}$

32. If the radius of first orbit of hydrogen atom is a_0, which of the following cannot be the radius of any other orbits of hydrogen?
(A) $4a_0$ (B) $6a_0$
(C) $9a_0$ (D) $16a_0$

33. Which of the following species has the largest bond angle?
(A) NH_4^+ (B) NO_2^+
(C) NO_2^- (D) NO_3^-

34. What is the oxidation number of iodine in KI_3?
(A) -1 (B) +1
(C) $-\frac{1}{3}$ (D) $+\frac{1}{3}$

Space for Rough work

35. $Cu^{2+} + 2e^- \longrightarrow Cu,\ E^o = +0.34\,V$

$Cu^{2+} + e^- \longrightarrow Cu^+,\ E^o = +0.15\,V$

The equilibrium constant for the reaction $Cu + Cu^{2+} \longrightarrow 2Cu^+$ at 25°C

(A) 2.9×10^{-8} (B) 3.72×10^{-7}

(C) 2.9×10^{-5} (D) 3.72×10^{-5}

36. At high pressure the van der Waal's equation for one mole of a real gas reduces to

(A) $PV = RT - \frac{a}{V}$ (B) PV = RT + Pb

(C) $\left(P + \frac{a}{V^2}\right)V = RT$ (D) PV = RT

37. 100 mL of a sample of hard water requires 25.1 mL of 0.02 N H_2SO_4 for complete reaction. The hardness of water (density 1 g/mL) is:

(A) 200 ppm (B) 250 ppm

(C) 251 ppm (D) 258 ppm

38. Two grams of dihydrogen diffuse from a container in 10 minutes. How many gram of dioxygen would diffuse through the same container in the same time under similar conditions?

(A) 4 (B) 8

(C) 6 (D) 0.5

39. The electronegativity of phosphorus is maximum in

(A) PCl_5 (B) $PClF_4$

(C) PCl_2F_3 (D) PCl_3F_2

40. A one litre vessel contains 3 moles each of gases A, B, C and D at equilibrium. If one mole each of A and B are removed, K_C for $A(g) + B(g) \rightleftharpoons C(g) + D(g)$ will be

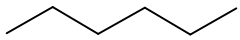
(A) 2.5 (B) 4

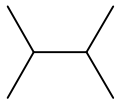
(C) 1 (D) ¼

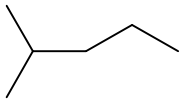
Space for Rough work

41. $CH_3CH_2CH_2Cl + CH_3\underset{\underset{Cl}{|}}{C}HCH_3 \xrightarrow[\text{Dry ether}]{Na} Products$

Which of the following is not a product of above reaction?

(A) 

(B) 

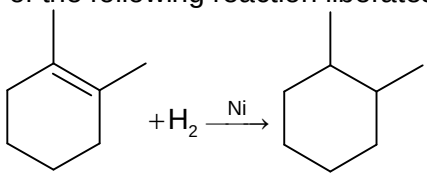
(C) 

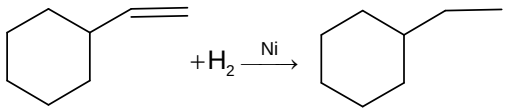
(D) 

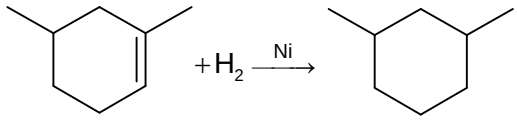
42. The half-life of a chemical reaction does not change by either increasing or decreasing the initial concentration of reactant. What is the order of the reaction?
(A) Zero (B) First
(C) Second (D) Fractional

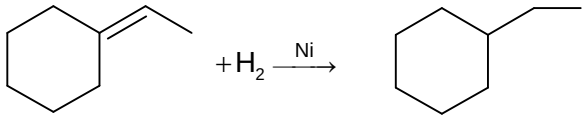
43. K_{sp} of AgCl in water at 25°C is 1.8×10^{-10}. If 10^{-5} mole of Ag^+ ions are added to this solution, K_{sp} will be:
(A) 1.8×10^{-15} (B) 1.8×10^{-5}
(C) 1.8×10^{-10} (D) none of these

44. Which of the following reaction liberates maximum amount of heat energy?

(A) 

(B) 

(C) 

(D) 

Space for Rough work

45. A diene on reductive ozonolysis produces two moles of ethanal and one mole of propan -1, 3-dial. How many geometrical isomer(s) is/are possible for the diene?
(A) 2 (B) 3
(C) 4 (D) 6

46. How much work in kJ mol^{-1} unit is done by reversible and isothermal expansion of 1.2 mole of an ideal gas to 10 times of its original volume at 27^oC?
(A) 4.191 (B) 6.892
(C) -6.892 (D) -4.191

47. In which of the following alcohols C – O bond breaks (heterolytically) with greatest ease:
(A) $CH_3CH_2CH_2OH$ (B) $CH_3-CH(CH_3)-OH$
(C) $CH_3-C(CH_3)_2-OH$ (D) $CH_3-C(CH_3)_2-CH_2-OH$

48. Which of the following has the most acidic hydrogen?
(A) 3-hexanone (B) 2, 4-hexanedione
(C) 2, 5-hexanedione (D) 2, 3-hexanedione.

49. Dehydration of 1-butanol gives 2-butene as a major product, by which of the following intermediate the compound 2-butene obtained.
(A) $H_3C-CH_2-CH_2^+$ (B) $H_3C-C^+(CH_3)-CH_2$
(C) $H_3C-CH^+-CH_3$ (D) $H_3C-CH(CH_3)-CH_2^+$

Space for Rough work

50. In which of the following crystals, alternate tetrahedral voids are occupied?
(A) NaCl (B) ZnS
(C) CaF_2 (D) Na_2O

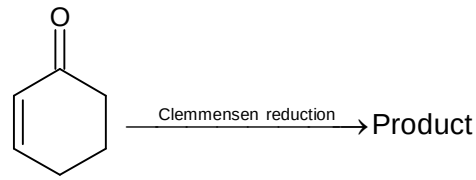
51. Molar heat capacity of an ideal gas at constant volume is given by $C_v = 2 \times 10^{-2}$ J (in Joule). If 3.5 mole of this ideal gas are heated at constant volume from 300 K to 400 K the change in internal energy will be
(A) 7 (B) 4
(C) 8 (D) 6

52. $BCl_3 + H_2 + C \xrightarrow{1700-1800^oC} Product(s)$
The product(s) of above reaction is/are
(A) $CCl_4 + B_2H_6$ (B) $B_4C + HCl$
(C) $B + Cl_2 + CH_4$ (D) $B_2H_6 + CHCl_3$

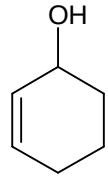
53. Atomic radii of Ti, Zr and Hf vary
(A) Ti > Zr > Hf (B) Ti < Zr < Hf
(C) Ti < Hf < Zr (D) Ti < Zr = Hf

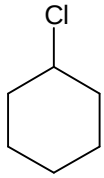
54. $C_6H_5CHO + CH_3COOCOCH_3 \xrightarrow[\Delta]{CH_3COONa} Products$
If one product of above reaction is $C_6H_5 - CH = CH - COOH$, the other organic product(side product) is:
(A) CH_3CH_2OH (B) C_6H_5COOH
(C) CH_3CHO (D) $C_6H_5 - CH = CH_2$

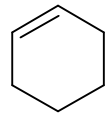
Space for Rough work

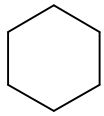
55. 

The product of above reaction is:

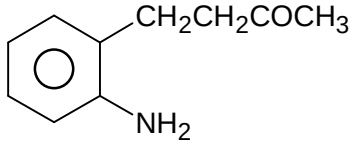
(A) 

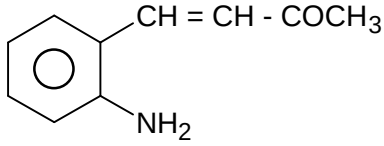
(B) 

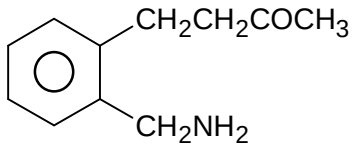
(C) 

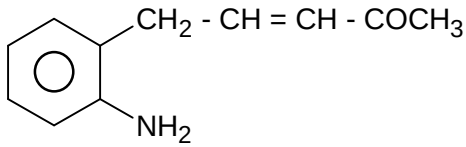
(D) 

56. Which forms aromatic compound on heating?

(A) 

(B) 

(C) 

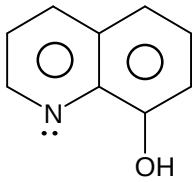
(D) 

57. The shape of $[Co(NH_3)_6]^{3+}$ is:
(A) Tetrahdedral (B) Trigonal bipyramidal
(C) Octahedral (D) Square planar

58. $AgNO_3$ solution will not give white precipitate with:
(A) $[Co(NH_3)_3Cl_3]$ (B) $[Co(NH_3)_4Cl_2]Cl$
(C) $[Co(NH_3)_5Cl]Cl_2$ (D) $[Co(NH_3)_6]Cl_3$

Space for Rough work

59. Which of the following two reagents form silicates when react with silica (SiO_2)?
(A) NaCl and Na_2S (B) NaOH and Na_2CO_3
(C) Na_2SO_4 and $NaNO_3$ (D) NaBr and $NaHSO_4$

60. In which combinations, all the ligands are ambient in nature?
(A) NO_2^-, $\bar{S}CN$ and $\bar{C}N$ (B) EDTA, $\bar{S}CN$
(C) H_2O,  (D) All of these

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$ has:
(A) no solution for x
(B) exactly one solution for x
(C) exactly two distinct solutions for x
(D) exactly three distinct solutions for x

62. The value of x satisfying the equation, $x = \sqrt{2+\sqrt{2-\sqrt{2+x}}}$ is:
(A) $2\cos 10^o$
(B) $2\cos 20^o$
(C) $2\cos 40^o$
(D) $2\cos 80^o$

63. If α, β are two real numbers satisfying $\alpha^2 + \beta^2 = 5$ and $3(\alpha^5 + \beta^5) = 11(\alpha^3 + \beta^3)$, then $\alpha\beta$ is:
(A) 2
(B) 1
(C) 7
(D) 9

64. $\frac{1}{4} + \frac{1(3)}{4(6)} + \frac{1(3)(5)}{4(6)(8)} + ...\infty$
(A) 1
(B) 2
(C) 3
(D) 4

Space for Rough work

65. For $x \in (1,\infty)$, if $P = x^3 - \frac{1}{x^3}$ and $Q = x - \frac{1}{x}$, then the minimum value of $\frac{P}{Q^2}$ equals:

(A) $2\sqrt{3}$ (B) $4\sqrt{3}$

(C) $\sqrt{3}$ (D) $3\sqrt{3}$

66. If the value of $\alpha \in [0,2\pi]$ such that the inequality $\sin\left(\frac{\pi}{3}+x\right)+\sin(\alpha+x) \geq 0$ is true for all real numbers x, is equal to $\frac{p\pi}{q}$ where p and q are relatively prime positive integers, then the value of $(p+q)$ is

(A) 7 (B) 6

(C) 5 (D) 4

67. $\underset{N\to\infty}{\text{Lim}}\ \underset{M\to\infty}{\text{Lim}} \sum_{R=1}^{N}\sum_{K=1}^{MR} \frac{MN^2}{\left(M^2N^2+K^2\right)\left(N^2+R^2\right)} = ?$

(A) $\frac{\pi^2}{32}$ (B) $\frac{\pi^2}{16}$

(C) $\frac{\pi^2}{25}$ (D) $\frac{\pi^2}{8}$

68. Number of straight lines passing through the point P (4, 3) whose x – intercept is a prime number and whose y – intercept is a positive integer, is equal to:

(A) 0 (B) 1

(C) 2 (D) more than 2 but finite

69. If a circle of radius 3 units is touching the lines $\sqrt{3}y^2 - 4xy + \sqrt{3}x^2 = 0$ in the first quadrant then the length of chord of contact to this circle is:

(A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{\sqrt{3}+1}{\sqrt{2}}$

(C) $3\left(\frac{\sqrt{3}+1}{\sqrt{2}}\right)$ (D) $3\frac{\left(\sqrt{3}+1\right)}{2}$

Space for Rough work

70. Identify the correct statement
(A) If system of n simultaneous linear equation has a unique solution, then coefficient matrix is singular
(B) If system of n simultaneous linear equation has a unique solution, then coefficient matrix is non singular
(C) If A^{-1} exists, $(adjA)^{-1}$ may or may not exist
(D) $F(x)=\begin{bmatrix}\cos x & -\sin x & 0\\ \sin x & \cos x & 0\\ 0 & 0 & 0\end{bmatrix}$, then $F(x).F(y)=F(x-y)$

71. A four digit number is called a doublet if any of its digit is the same as only one neighbour. For example, 1221 is a doublet but 1222 is not. Number of such doublets are:
(A) 2259 (B) 2268
(C) 2277 (D) 2349

72. The constant term in the expansion of $\left(x^2+\frac{1}{x^2}+y+\frac{1}{y}\right)$ is:
(A) 4900 (B) 4950
(C) 5050 (D) 5151

73. The coefficient of x^{11} in the expression $(1+x)^5(3+x)^4(7+x)^3$ equals:
(A) 28 (B) 34
(C) 38 (D) 42

74. The function $f:R\to R$ defined as $f(x)=\ln\left(\sqrt{\sqrt{x^2+1}+x}+\sqrt{\sqrt{x^2+1}-x}\right)$ is:
(A) one – one and onto both (B) one – one but not onto
(C) onto but not one – one (D) neither one – one nor onto

Space for Rough work

75. Let $f(x)=\ln x$ and $g(x)=\dfrac{x^4-2x^3+3x^3-2x+2}{2x^2-2x+1}$. The domain of the composite function $fog(x)$ is:

(A) $(-\infty, \infty)$ (B) $[0,\infty)$

(C) $(0, \infty)$ (D) $[1, \infty)$

76. Let $S_n=\cot^{-1}\left(3x+\dfrac{2}{x}\right)+\cot^{-1}\left(6x+\dfrac{2}{x}\right)+\cot^{-1}\left(10x+\dfrac{2}{x}\right)+.....+n$ terms, where $x>0$. If $\lim\limits_{n\to\infty} S_n=1$, then x equals

(A) $\dfrac{\pi}{4}$ (B) 1

(C) $\tan 1$ (D) $\cot 1$

77. $\lim\limits_{n\to\infty}\left(\sqrt{n^2+n+1}-\left[\sqrt{n^2+n+1}\right]\right)(n\in I)$ where $[.]$ denotes the greatest integer function is:

(A) 0 (B) $\dfrac{1}{2}$

(C) $\dfrac{2}{3}$ (D) $\dfrac{1}{4}$

78. Let $f:R\to R$ be a continuous onto function satisfying $f(x)+f(-x)=0, \forall x\in R$. If $f(-3)=2$ and $f(5)=4$ in $[-5, 5]$, then the equation $f(x)=0$ has:

(A) exactly three real roots (B) exactly two real roots

(C) at least five real roots (D) at least three real roots

79. Let f be differentiable at x = 0 and $f'(0)=1$. Then $\lim\limits_{h\to 0}\dfrac{f(h)-f(-2h)}{h}=$

(A) 3 (B) 2

(C) 1 (D) –1

Space for Rough work

80. Let $f(x)=\begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$ where p is a constant. Then $\frac{d^3(f(x))}{dx^3}$ at $x=0$ is:

(A) $6p^3$ (B) $p+p^2$

(C) $p+p^3$ (D) independent of p

81. If $\int u\frac{d^2v}{dx^2}dx = u\frac{dv}{dx} - v\frac{du}{dx} + w$, then w is equal to:

(A) $\int v\frac{d^2u}{dx^2}dx$ (B) $\int u\frac{d^2v}{dx^2}dx$

(C) $\int v\left(\frac{du}{dx}\right)^2 dx$ (D) $\int u\left(\frac{dv}{dx}\right)^2 dx$

82. $\lim\limits_{n\to\infty}\frac{n}{(n!)^{1/n}}$ is equal to

(A) e (B) $\frac{1}{e}$

(C) 1 (D) $\int\limits_0^1 \ln x\,dx$

83. $\int\limits_0^1 x\ln\left(1+\frac{x}{2}\right)dx =$ _____

(A) $\frac{3}{4}\left(1-2\ln\frac{3}{2}\right)$ (B) $\frac{3}{2}-\frac{7}{2}\ln\frac{3}{2}$

(C) $\frac{3}{4}+\frac{1}{2}\ln\frac{1}{54}$ (D) $\frac{1}{2}\ln\frac{27}{2}-\frac{3}{4}$

Space for Rough work

84. Number of roots of the equation $x^2.e^{2-|x|}=1$ is:

(A) 2 (B) 4

(C) 6 (D) zero

85. A curve passing through (2, 3) and satisfying the differential equation $\int_0^x t\,y(t)\,dt = x^2 y(x), (x>0)$ is:

(A) $x^2+y^2=13$ (B) $y^2=\frac{9}{2}x$

(C) $\frac{x^2}{8}+\frac{y^2}{18}=1$ (D) $xy=6$

86. The line $y=mx$ bisects the area enclosed by the curve $y=1+4x-x^2$ and the lines $x=0,\ x=\frac{3}{2}$ and $y=0$. Then the value of m is:

(A) $\frac{13}{6}$ (B) $\frac{6}{13}$

(C) $\frac{3}{2}$ (D) 4

87. Let $N=\begin{vmatrix}28 & 25 & 38\\ 42 & 38 & 65\\ 56 & 47 & 83\end{vmatrix}$, then the number of ways in which N can be roots as a product of two divisors which are relatively prime is:

(A) 4 (B) 8

(C) 9 (D) 16

Space for Rough work

88. The value of m for which straight line $3x-2y+z+3=0=4x-3y+4z+1$ is parallel to the plane $2x-y+mz-2=0$ is

(A) –2 (B) 8
(C) –18 (D) 11

89. For the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$, which one of the following is incorrect?

(A) it lies in the plane $x-2y+z=0$

(B) it is same as line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) it passes through (2, 3, 5)

(D) it is parallel to the plane $x-2y+z-6=0$

90. The graph of the equation $x+y=x^3+y^3$ is the union of:

(A) line and an ellipse (B) line and a parabolas
(C) line and hyperbola (D) line and a point

Space for Rough work